

ENGINEERING EXPLORATION



What is Engineering



- Engineering is the practical application of mathematics, science, and technology to design, create, and maintain systems, structures, and processes.
- It is problem-solving at its core, aimed at turning scientific knowledge into practical, real-world solutions that improve human life

Science vs Engineering



Science focuses on expanding human knowledge by discovering and explaining the natural world, while engineering focuses on applying that knowledge to design and build practical solutions to human problems.

Scientist vs Engineer

Scientist	Engineer
• Discovers how nature works. • Asks “Why?” and “What happens?” • Focuses on theories, experiments, and research. • Creates new knowledge.	• Uses scientific knowledge to solve problems. • Asks “How can we build or improve it?” • Focuses On design, construction, and implementation. • Creates useful products and systems.

Needs and wants of computer science in today's world



Needs of Computer Science

Needs are essential uses of computer science that help society function effectively.

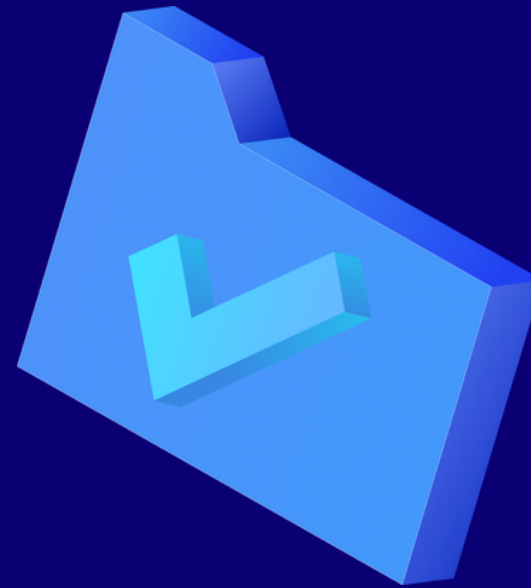
- Communication – Enables email, messaging, and video conferencing.
- Education – Supports online learning, digital classrooms, and e-books.
- Healthcare – Helps maintain patient records, diagnosis, and telemedicine.
- Business – Manages accounts, inventory, payroll, and online transactions.
- Transportation – Assists in traffic management, GPS navigation, and ticket booking.
- Cybersecurity – Protects data and systems from cyber threats.

Wants of Computer Science

Wants are additional features and advancements that improve convenience and quality of life.

- Artificial Intelligence (AI) – Smart assistants, chatbots, and automation.
- Cloud Computing – Access data and applications from anywhere.
- Smart Homes – Control lights, security, and appliances using the internet.
- Entertainment – Streaming movies, music, and online gaming.
- Social Media – Connect and share information globally.
- Robotics – Used in industries, healthcare, and space exploration.

DATA MANAGEMENT



DATABASE

A project is a carefully organized effort, often with a specific goal or set of objectives in mind, designed to address a particular problem or opportunity. In professional environments, projects are essential tools for driving innovation, streamlining processes, or creating new products and services.



DATA SECURITY

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DATA ANALYSIS

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Miscomsuption about CSE

Misconception: CSE students only do coding.

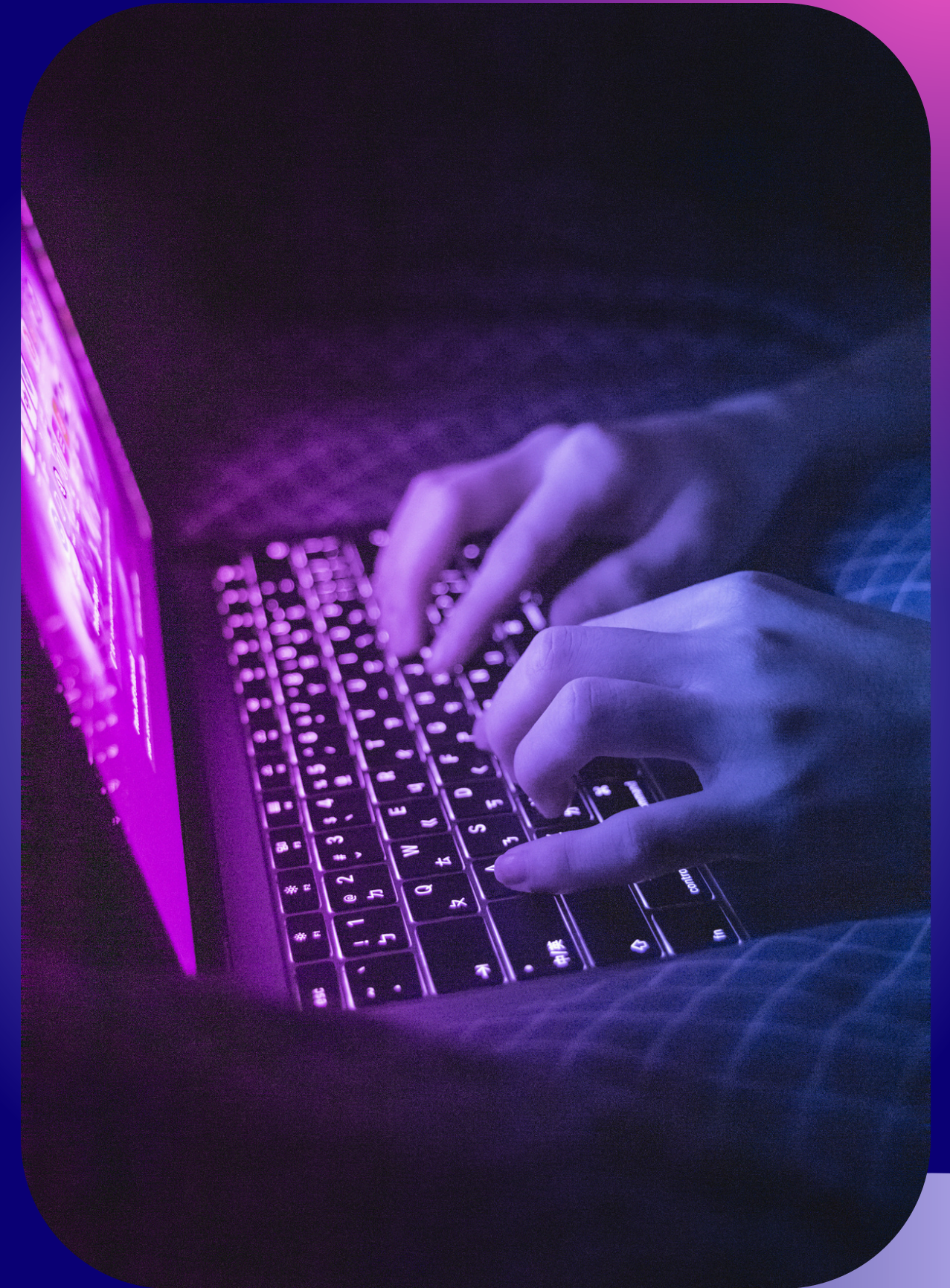
Reality: They also learn algorithms, databases, networking, artificial intelligence, cybersecurity, cloud computing, and software engineering.

Misconception: Programming is the only skill needed.

Reality: Employers also value communication, teamwork, creativity, critical thinking, and problem-solving skills.

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SKILLS REQUIRED FOR COMPUTER SCIENCE AND ENGINEERING (CSE) IN THE 21ST CENTURY

Programming Skills

- Knowledge of programming languages like C, C++, Java, Python, and JavaScript.
- Ability to write efficient and error-free code.

Problem-Solving Skills

- Ability to analyze problems and find logical solutions.
- Strong analytical and critical thinking

Artificial Intelligence and Machine Learning

- Basic knowledge of AI, ML, and data science.
- Ability to develop intelligent applications.

Communication Skills

- Ability to explain ideas clearly.
- Good verbal and written communication.

Teamwork and Leadership

- Ability to work effectively in a team.
- Leadership skills to manage projects and guide team members.

Cybersecurity

- Understanding of network security, ethical hacking, and data protection.
- Awareness of cyber threats and prevention methods.

Thank You